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Module 4 – Lecture 13

Processes, information, martingales and stopping times

A process, a random variable and a path

Definition

On one probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a stochastic process is a family $(X_t)_{t \in T}$ of random variables with values in a state space E.

- $T = \mathbb{N}_0$: discrete time; $T = [0, \infty)$: continuous time.
- Fix t: X_t is a random variable.
- Fix ω: $t \mapsto X_t(\omega)$ is a sample path.
- Fix both: $X_t(\omega)$ is a value in E.

One example throughout: the simple random walk

Let (ξ_k) be independent with $\mathbb{P}(\xi_k = 1) = \mathbb{P}(\xi_k = -1) = 1/2$. Define

$$S_0 = 0, \qquad S_n = \sum_{k=1}^{n} \xi_k.$$

- Time: $\mathbb{N}_0$. State space: $\mathbb{Z}$.
- One path is one sequence of cumulative outcomes.
- $\mathbb{E}[S_n] = 0$, but $\mathrm{Var}(S_n) = n$.

Quick check

What are the possible values of S_2? What are their probabilities?

Finite-dimensional distributions

The joint laws of $(X_{t_1}, \ldots, X_{t_m})$, for every finite choice of times, describe dependence across time.

Law as a random path

The map $X : \omega \mapsto (X_t(\omega))_{t \in T}$ induces a probability law on E^T with its product sigma-algebra. All finite-dimensional distributions determine that law.

For continuous paths on $[0, T]$, we can instead use $C([0, T])$ with its Borel sigma-algebra. We will return to this for Brownian motion.

Equal marginals do not give equal joint laws

Let Z, Z_1, Z_2 be independent standard normal variables. Set

$$(X_1, X_2) = (Z, Z), \qquad (Y_1, Y_2) = (Z_1, Z_2).$$

- Each of the four variables has law $N(0, 1)$.
- But $\mathrm{Cov}(X_1, X_2) = 1$ and $\mathrm{Cov}(Y_1, Y_2) = 0$.

Conclusion

One-time distributions do not determine a process law. We need joint distributions.

Measurability: three different questions

- For each t, is $\omega \mapsto X_t(\omega)$ measurable? This is part of the definition of a process.
- Is $(t, \omega) \mapsto X_t(\omega)$ jointly measurable from $\mathcal{B}([0, T]) \otimes \mathcal{F}$ to $\mathcal{B}(\mathbb{R})$? This defines a measurable process.
- For each t, is X_t measurable using only information available by time t? This will define adaptedness.

Useful sufficient condition

A real-valued process with continuous paths is jointly measurable. We use standard continuous or right-continuous versions in our examples.

Stationarity versus stationary increments

- Strict stationarity: all joint laws are unchanged by a common time shift.
- Weak stationarity: finite second moments, constant mean, and covariance depending only on the lag.
- Stationary increments: the law of $X_{t+h} - X_t$ depends only on h.
- Independent increments: increments over disjoint time intervals are independent.

Random walk

S_n has stationary independent increments, but is not weakly stationary because $\mathrm{Var}(S_n) = n$.

Filtrations represent information over time

Definition

A filtration $(\mathcal{F}_t)$ is an increasing family of sub-sigma-algebras of $\mathcal{F}$:

$$\mathcal{F}_s \subseteq \mathcal{F}_t \qquad (s \leq t).$$

An event in $\mathcal{F}_t$ can be decided using the information available at time t.

Natural filtration

$$\mathcal{F}_t^X = \sigma(X_u : 0 \leq u \leq t).$$

For the random walk, $\mathcal{F}_n^S = \sigma(\xi_1, \ldots, \xi_n)$.

Adapted means observable now

Definition

X is adapted to $(\mathcal{F}_t)$ if X_t is $\mathcal{F}_t$-measurable for every t.

Random walk

$S_n = \xi_1 + \cdots + \xi_n$ is $\mathcal{F}_n$-measurable. Hence S is adapted to $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$.

Using future information

$Y_n = S_{n+1}$ is not adapted to this filtration: it uses the new, nondegenerate shock ξ_{n+1}.

Predictable means chosen before the next shock

Discrete-time definition

H_n is predictable if it is $\mathcal{F}_{n-1}$-measurable for each $n \geq 1$.

- $H_n = \mathbf{1}_{\{S_{n-1}>0\}}$ is predictable.
- $H_n = S_n$ is adapted but not predictable for the simple random walk.

Trading interpretation

In $H_n(S_n - S_{n-1})$, the holding H_n is chosen before the increment is observed.

Continuous-time predictability will be introduced with stochastic integration.

Checkpoint: which information is being used?

Discuss before revealing the next slide

For the simple random walk with $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$, classify:

$$A_n = S_n, \qquad H_n = \mathbf{1}_{\{S_{n-1} \geq 0\}}, \qquad Y_n = S_{n+1}.$$

Which are adapted? Which are predictable for $n \geq 1$?

Also explain why constant mean alone does not make S_n stationary.

Checkpoint: answers

- $A_n = S_n$: adapted, not predictable.
- $H_n = \mathbf{1}_{\{S_{n-1} \geq 0\}}$: predictable and therefore adapted.
- $Y_n = S_{n+1}$: not adapted and not predictable.
- $\mathbb{E}[S_n] = 0$ is constant, but $\mathrm{Var}(S_n) = n$ is not.

End of Block 1

A process needs joint laws; information must be specified before discussing conditional predictions.

Conditional expectation: the rules we need

For an integrable random variable Y, $\mathbb{E}[Y \mid \mathcal{F}_s]$ is an $\mathcal{F}_s$-measurable conditional prediction.

- If Y is $\mathcal{F}_s$-measurable, then $\mathbb{E}[Y \mid \mathcal{F}_s] = Y$.
- If Y is independent of $\mathcal{F}_s$, then $\mathbb{E}[Y \mid \mathcal{F}_s] = \mathbb{E}[Y]$.
- Known factors can be taken outside, when the products are integrable.
- Tower property: $\mathbb{E}[\mathbb{E}[Y \mid \mathcal{F}_t] \mid \mathcal{F}_s] = \mathbb{E}[Y \mid \mathcal{F}_s]$ for $s \leq t$.

For square-integrable Y, this is also the best mean-square predictor using $\mathcal{F}_s$.

Martingales: three conditions

Definition

A process (M_t) is a martingale with respect to $(\mathcal{F}_t)$ if

1. M_t is $\mathcal{F}_t$-measurable;
2. $\mathbb{E}[|M_t|] < \infty$ for every t;
3. $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$ for all $s \leq t$.

All conditional equalities hold almost surely.

Meaning

Given current information, the expected future value equals the current value. A martingale need not have independent increments.

Worked proof: the random walk is a martingale

S_n is adapted and $|S_n| \leq n$, so it is integrable. Also

$$S_{n+1} = S_n + \xi_{n+1}.$$

Since ξ_{n+1} is independent of $\mathcal{F}_n$ and centered,

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n + \mathbb{E}[\xi_{n+1}] = S_n.$$

From one step to many steps

Repeated conditioning and the tower property give the martingale equality for every pair of times $m \leq n$.

Drift, compensation and inequality versions

If the increments are i.i.d. and integrable with mean μ,

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n + \mu.$$

- $\mu = 0$: S is a martingale.
- $\mu \geq 0$: S is a submartingale.
- $\mu \leq 0$: S is a supermartingale.

Remove the predictable mean

In every case, $M_n = S_n - n\mu$ is a martingale.

In general: submartingale means $\mathbb{E}[X_t \mid \mathcal{F}_s] \geq X_s$; supermartingale means the reverse. Both require adaptedness and integrability.

Exercise: the square of the random walk

Work for 5 minutes

For the simple symmetric random walk, show that

$$M_n = S_n^2 - n$$

is a martingale. Is S_n^2 itself a martingale?

- Check adaptedness and integrability.
- Expand $(S_n + \xi_{n+1})^2$.
- Use $\mathbb{E}[\xi_{n+1}] = 0$ and $\xi_{n+1}^2 = 1$.

Solution: the correction is the accumulated variance

M_n is adapted and integrable, since $|S_n| \leq n$. Moreover,

$$S_{n+1}^2 = S_n^2 + 2S_n\xi_{n+1} + \xi_{n+1}^2.$$

Conditioning gives

$$\mathbb{E}[S_{n+1}^2 \mid \mathcal{F}_n] = S_n^2 + 1.$$

Therefore

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = S_n^2 + 1 - (n+1) = M_n.$$

Interpretation

S_n^2 is a submartingale. Subtracting n removes its conditional growth.

A predictable strategy preserves the martingale property

Let M be a discrete-time martingale and let H_k be bounded and $\mathcal{F}_{k-1}$-measurable. Set

$$G_n = \sum_{k=1}^{n} H_k (M_k - M_{k-1}).$$

Then G is a martingale because

$$\mathbb{E}[G_n - G_{n-1} \mid \mathcal{F}_{n-1}] = H_n \mathbb{E}[M_n - M_{n-1} \mid \mathcal{F}_{n-1}] = 0.$$

One example

$H_k = \mathbf{1}_{\{S_{k-1} \geq 0\}}$ is a valid bounded predictable strategy for the simple random walk.

Stopping times: decisions that do not look ahead

Definition

A random time $\tau : \Omega \to [0, \infty]$ is a stopping time if

$$\{\tau \leq t\} \in \mathcal{F}_t \qquad \text{for every } t.$$

At time t, we can decide whether stopping has already occurred.

- A deterministic time is a stopping time.
- A first observed threshold crossing is a standard example.
- A rule based on the last crossing before a future deadline generally looks ahead.

First hitting time: a complete discrete-time proof

Let X_n be adapted and set

$$\tau = \inf\{n \geq 0 : X_n \geq a\}, \qquad \inf \varnothing = \infty.$$

For each n,

$$\{\tau \leq n\} = \bigcup_{k=0}^{n} \{X_k \geq a\}.$$

Each event belongs to $\mathcal{F}_k \subseteq \mathcal{F}_n$. The finite union belongs to $\mathcal{F}_n$. Hence τ is a stopping time.

Question

Why does this proof use first crossing rather than last crossing?

A concrete random time that is not a stopping time

For the simple random walk observed up to time 2, let

$$L = \max\{k \in \{0, 1, 2\} : S_k = 0\}.$$

This is well-defined because $S_0 = 0$. Since $S_1 \neq 0$,

$$\{L \leq 1\} = \{S_2 \neq 0\} = \{\xi_2 = \xi_1\}.$$

Why this is not known at time 1

Given either value of ξ_1, the event still has conditional probability $1/2$. It is not $\mathcal{F}_1$-measurable, so L is not a stopping time.

Optional sampling: a safe discrete-time statement

Bounded stopping times

If (M_n) is a discrete-time martingale and $\tau \leq N$ is a stopping time for a fixed finite N, then

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_0].$$

Application

For $\tau = \inf\{n : S_n = 2\}$ and $\sigma = \tau \wedge 10$,

$$\mathbb{E}[S_\sigma] = 0.$$

The bounded theorem applies to σ, not automatically to τ.

Stopping without future information does not by itself replace the hypotheses of the theorem.

Guided exercise: information, martingales and stopping

Work for 8 minutes

Let S be the simple random walk and $\tau = \inf\{n \geq 0 : |S_n| = 2\}$. Set $\sigma = \tau \wedge 4$.

1. Show that τ and σ are stopping times.
2. Compute $\mathbb{E}[S_\sigma]$.
3. Use $S_n^2 - n$ to relate $\mathbb{E}[S_\sigma^2]$ and $\mathbb{E}[\sigma]$.

You do not need the exact distribution of τ.

Guided exercise: solution

- $\{\tau \le n\} = \bigcup_{k=0}^{n} \{|S_k| = 2\} \in \mathcal{F}_n$.
- $\{\sigma \le n\} = \{\tau \le n\}$ if $n < 4$, and Ω if $n \ge 4$.
- Since $\sigma \le 4$, optional sampling gives $\mathbb{E}[S_\sigma] = \mathbb{E}[S_0] = 0$.
- Apply the same theorem to $M_n = S_n^2 - n$:

$$\mathbb{E}[S_\sigma^2 - \sigma] = 0, \qquad \mathbb{E}[S_\sigma^2] = \mathbb{E}[\sigma].$$

What made this possible?

A specified filtration, a stopping-time rule, and a bounded stopping time.

What you should be able to do after Lecture 13

- Distinguish a process, a path, marginals and joint laws.
- Distinguish joint measurability, adaptedness and discrete predictability.
- Verify a martingale by checking all three conditions.
- Recognize first hitting times and detect rules using the future.
- Apply bounded discrete optional sampling in simple examples.

Next lecture

We apply these ideas to Brownian and Poisson noise, then describe Markov evolution using transitions and generators.